Multiple Choice:

1. If the coefficient of sliding friction for steel on ice is 0.05, what force is required to keep a man weighing 150 lbs moving at constant speed along the ice?

A. 7.5 lb C. 15 lb
B. 12.5 lb D. 20 lb

2. A 400 kN block is resting on a rough horizontal surface for which the coefficient of friction is 0.40. Determine the force P required to cause motion to impend if applied to the block.

A. 140 kN C. 180 kN
B. 160 kN D. 200 kN

3. A trapezoid with parallel bases measure 2 m and 4 m, respectively. Find the location of its center of gravity, measured from the longer side, if its altitude is 3 m.

A. 1.21 C. 1.33
B. 1.27 D. 1.41

4. A 600 N block rests on a 30° plane. If the coefficient of static friction is 0.30 and the coefficient of kinetic friction is 0.20, what is the value of the force, P applied horizontally to keep the block moving up the plane at constant speed?

A. 459 N C. 515 N
B. 505 N D. 527 N

5. An inverted cone (apex down) has a mass of 10. If its base (on top) has a radius of 2 and its height is 3, find the moment of inertia of the cone with respect to its vertical axis passing through the center of the base.

A. 12 C. 18
B. 15 D. 22

6. A 65-lb horizontal force is sufficient to draw a 1200-lb sled on level, well-packed snow at uniform speed. What is the value of the coefficient of friction?

A. 0.054 C. 0.156
B. 0.102 D. 0.298

7. The block of weight W is supported by a rope that is wrapped one and one half times around the circular peg. Determine the range of values of P for which the block remains at rest. (μ_s=0.2)

A. 0.152W £ P £ 6.59W
B. 0.192W £ P £ 3.67W
C. 0.138W £ P £ 2.30W
D. 0.217W £ P £ 5.81W

8. A crate of mass 100kg rests on the floor. The coefficient of static friction is 0.4. If a force of 250 N (parallel to the floor) is applied to the crate, what's the magnitude of the force of static friction on the crate?

A. 400 N C. 250 N
B. 300 N D. 200 N

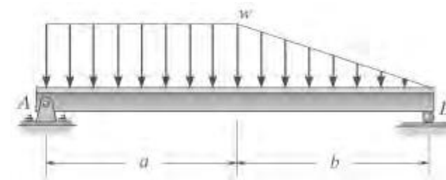
9. The crate in an inclined plane (angle of inclination, θ_2=10°), has a mass M=200 kg and is subjected to a towing force P acting at an angle θ_1=20° with the horizontal. If the coefficient of static friction is 0.2, determine the magnitude of P to just start the crate moving down the plane.

A. N=1658 N, P=37 N
B. N=1909 N, P=47 N
C. N=1780 N, P=52 N
D. N=1687 N, P=39 N

10. The 100-lb block is at rest on a rough horizontal plane before the force P is applied. Determine the magnitude of P that would cause impending sliding to the right. Given that μ_k=0.2, μ_s=0.3

A. 50 lb C. 30 lb
B. 40 lb D. 20 lb

11. If the maximum intensity of the distributed load acting on the beam is w, determine the reactions at the pin A and roller B. Given: a=3m, b=3m, w=4 kN/m



A. A_x=0 lb, A_y=2750 lb, B_y=1000 lb
B. A_x=0 lb, A_y=3910 lb, B_y=1500 lb
C. A_x=0 lb, A_y=1930 lb, B_y=1093 lb
D. A_x=0 lb, A_y=1092 lb, B_y=1028 lb

12. The box has weight W=90 lb and rests on a tile floor for which the coefficient of friction is 0.25. If the man pushes on it in the direction θ=30°, determine the smallest magnitude of force F needed to move the box.

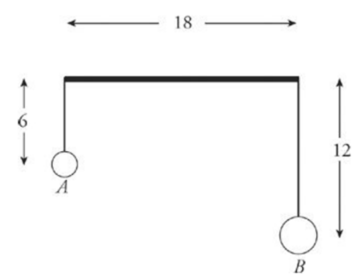
A. 30.4 lb C. 16.9 lb
B. 20.6 lb D. 10.2 lb

13. A block rests on a rough inclined plane making an angle of 30 degrees with the horizontal. The coefficient of static friction between the block and the plane is 0.8. If friction is 10 N, what is the mass of the block in kg?

A. 0.9 kg C. 1.5 kg
B. 1.2 kg D. 2 kg

14. Object A, of mass 5 kg, and object B, of mass 10 kg, hang from light threads from the ends of a uniform bar of length 18 and mass 15 kg. The masses A and B are at distances 6 and 12, respectively, below the bar. Find the center of mass of this system (assume that the 15 kg mass is at the origin).

A. (1.5, -5) C. (-1.5, -5)
B. (1.5, 5) D. (-1.5, 5)



15. A uniform chain of length 2 m is kept on a table such that a length of 60 cm hangs freely from the edge of the table. The total mass of the chain is 4 kg. What is the work done in pulling the entire chain on the table?

A. 3.6 J C. 1.5 kg
B. 1.2 kg D. 2 kg

16. A box is dragged up and down a concrete slope of 15° to the horizontal. To get the box started up the slope, it is necessary to exert six times the force needed to get it started down the slope. If the force is always parallel to the slope, what is the coefficient of static friction between the box and the concrete?

A. 0.264 C. 0.375
B. 0.294 D. 0.445

17. The rate of change in momentum is directly proportional to the impressed force, and takes place in the same direction in which the force acts. This statement is known as ______.

A. The Law of Inertia
B. The Law of Reaction
C. The Law of Acceleration
D. The Law of Conservation of Momentum

18. The principle of transmissibility of forces states that, when a force acts upon a body, its effect is ______.

A. the same at ever point on its line of action
B. minimum, if it acts at the center of gravity of the body
C. maximum, if it acts at the center of gravity of the body
D. different at different points on its line of action

19. Varignon's theorem of moments states that if a number of coplanar forces acting on a particle are in equilibrium, then _______.

A. the algebraic sum of their moments about any point is equal to the moment of their resultant force about the same point
B. their algebraic sum is zero
C. their line of action are at equal distance
D. the algebraic sum of their moments about any point in their plane is zero

20. Which of the following is NOT true about static friction?

A. It always acts in a direction, opposite to that in which the body tends to move
B. Is dependent on the area of contact, between the two surfaces
C. It bears a constant ratio to the normal reaction between the two surfaces
D. It is always greater than the kinetic friction

21. According to the Lami's Theorem:

A. The three forces must be equal
B. The three forces must be at 120 degrees to each other
C. If the three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two
D. If the three forces acting at a point are in equilibrium, then each force is inversely proportional to the sine of the angle between the other two

22. The acceleration of a body sliding down a smooth inclined surface is:

A. gsinθ C. gtanθ
B. gcosθ D. 1/gcosθ

23. How many reactions are present in the support of a cantilever beam?

A. 1 C. 3
B. 2 D. none

24. What is the magnitude of the equivalent concentrated load of a uniformly varying load applied to a beam of length L with maximum magnitude of w?

A. wL C. wL/3
B. wL/2 D. wL/4

25. A ladder is resting on a smooth ground and leaning against a rough vertical wall. The force of friction will act:

A. Downward at its upper end
B. Away from the wall at its upper end
C. Towards the wall at its upper end
D. Upward at its upper end

26. A 100-kg box is at rest on a horizontal surface. If the coefficient of static and kinetic friction are 0.4 and 0.3, respectively, and a force of 350 N parallel to the floor is applied to the box, which of the following is true?

A. The box will accelerate across the floor at 0.5 m/s^2
B. The box will not move
C. The box will slide across the floor at a constant speed of 0.5 m/s
D. The static friction force will have a value of 350 N

27. A 100-kg box is at rest on a horizontal surface. If the coefficient of static and kinetic friction are 0.4 and 0.3, respectively, and a force of 400 N parallel to the floor is applied to the box, which of the following is true?

A. The box is about to start to move
B. There will be a horizontal net force of 400 N – 392.4 N = 7.6 N
C. The box will slide across the floor with an acceleration of 1.06 m/s^2
D. The box will slide across the floor with an acceleration of 0.08 m/s^2

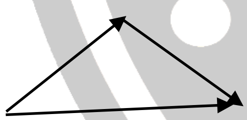
28. The frictional force is _______ the area of contact.

A. inversely proportional to
B. dependent on
C. independent on
D. directly proportional to

29. Two boxes with the same mass are stacked on top of each other on a horizontal surface. Box 1 is on the bottom and box 2 is on top of it. Compared with the strength of the force F_1 necessary to push only box 1 at constant speed across the floor, the force F_2 needed to push the 2 boxes across the floor at the same constant speed is greater than F_1 because _________.

A. the coefficient of static friction between box 1 and the floor is greater
B. the kinetic friction, but not the normal force, on box 1 is greater
C. the force of the floor on box 1 is greater
D. the coefficient of kinetic friction between box 1 and the floor is greater

30. Three forces acting on a body are as shown. Which diagram represents the body in equilibrium?

A.


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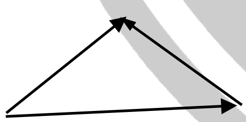

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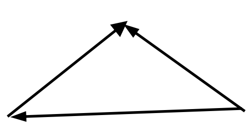
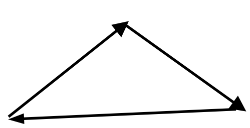
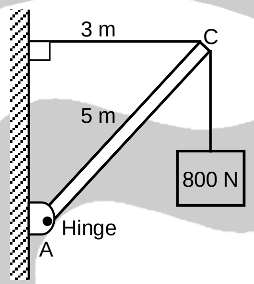


D.


31. If a ladder is leaning against a smooth wall at an angle θ with respect to the horizontal. If the ladder is in equilibrium, what is the coefficient of friction between the ladder and the ground?

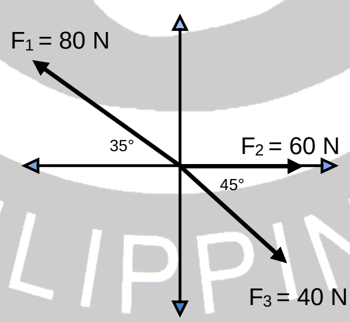
A. 0.5 tan θ　　C. tan θ
B. 0.5 cot θ　　D. cot θ

32. A 5.0-m weightless rod (AC), hinged to a wall at A, is used to support an 800-N block as shown in the figure. What is the total reaction in the hinge or pin?



A. 1000 N　　C. 1500 N
B. 1200 N　　D. 1800 N

33. Three force F_1, F_2, and F_3 act on a particle with mass of 3.80 kg as shown in the figure below. What should be the magnitude and direction of a fourth force that will make the system equilibrium?



A. 28.77 N at 217.7 degrees from +x
B. 34.55 N at 217.7 degrees from +x
C. 28.77 N at 183.3 degrees from +x
D. 34.55 N at 183.3 degrees from +x

34. Block A, with a mass of 10 kg, rests on a 35° incline. The coefficient of static friction is 0.40. An attached string is parallel to the incline and passes over a massless, frictionless pulley at the top. What is the largest mass of block B, attached to the dangling end, for which A begins to slide down the incline? Ans. 2.5 kg

A. 1.6 kg　　C. 2.3 kg
B. 1.8 kg　　D. 2.5 kg

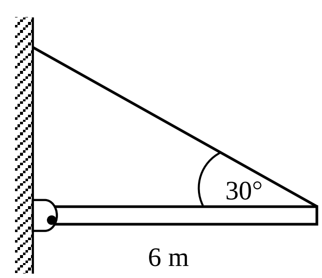
35. A uniform ladder of length L and mass M rests against a smooth, vertical wall. The coefficient of static friction between the ladder and the ground is 0.40. Find the angle θ (in degrees), between the ladder and the ground, at which the ladder is about to slide.

A. 51.3°　　C. 54.4°
B. 52.5°　　D. 55.1°

36. A block slides with constant velocity down an inclined plane 30 degrees with respect to the horizontal. What is the coefficient of kinetic friction between the block and the plane? If the block is projected up the plane with an initial speed of 2.5 m/s, how far up the plane will it move before coming to rest?

A. 0.45 m　　C. 0.58 m
B. 0.44 m　　D. 0.61 m

37. A horizontal uniform beam of weight W = 200 N and length L = 6.0 m is supported by a hinge and a cable as shown in the figure. The system is in equilibrium; find the tension in the cable.



A. 180 N C. 250 N
B. 200 N D. 300 N

38. A block of mass 2 kg slides on an inclined plane that makes a 30-degree angle with the horizontal. If the coefficient of friction between the block and the plane is 0.86, what should be the force parallel to the plane applied to the block so that it moves up the plane without any acceleration?

A. 21.6 N C. 24.4 N
B. 23.5 N D. 28.1 N

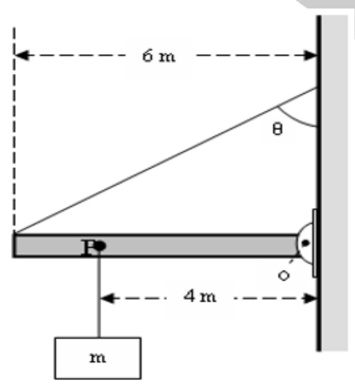
39. The friction experienced by a body, when in motion, is known as _____.

A. rolling friction
B. dynamic friction
C. limiting friction
D. static friction

40. The resultant of two equal forces has the same magnitude as either of the forces, then the angle between the two forces is:

A. 30 degrees
B. 60 degrees
C. 90 degrees
D. 120 degrees

41. A uniform horizontal beam of length 6.00 m and mass M = 100 kg is attached to the vertical wall by a rope making an angle θ = 60.0 ° with the wall; the beam can rotate about the fixed pivot O as shown in the figure. A mass m = 80.0 kg hangs from P at a distance 4.00 m from O. Find the tension in the rope attached to the wall.



A. 2 kN C. 4 kN
B. 3 kN D. 5 kN

42. Static friction is always _____ dynamic friction.

A. equal to
B. less than
C. greater than
D. less than or equal to

43. The maximum frictional force, which comes into play, when a body just begins to slide over the surface of the other body, is known as:

A. Maximum coefficient of friction
B. Dynamic friction
C. Limiting friction
D. Static friction

44. The algebraic sum of the resolved parts of a number of forces in a given direction is equal to the resolved part of their resultant in the same direction. This is known as ________.

A. Principle of independence of forces
B. Principle of resolution of forces
C. Principle of transmissibility of forces
D. None of the above

45. The angle between two forces when the resultant is maximum and minimum respectively are:

A. 0 and 180 degrees
B. 180 and 0 degrees
C. 90 and 180 degrees
D. 90 and 0 degrees

46. The motion of a particle round a fixed axis is ______.

A. translational
B. rotational
C. circular
D. both translational and circular

47. The resultant of the two forces P and Q is R. If Q is doubled, the new resultant is perpendicular to P. Then _____.

A. P = Q
B. Q = R
C. Q = 2R
D. None of the above

48. The static friction _______.

A. bears a constant ratio of the normal reaction between the two surfaces
B. is independent of the area of the contact between the two surfaces
C. always acts in a direction, opposite to that in which the body tends to move
D. all of the above

49. The angle of inclination of the plane at which the body begins to move down the plane is called _____.

A. angle of friction
B. angle of repose
C. angle of projection
D. none of the above

50. The ratio of static friction to dynamic friction is always:

A. Equal to 1
B. Less than 1
C. Greater than 1
D. Less than or equal to 1